PART A — MODULE 1 QUESTIONS

MATH3-M1-001
Prompt:
What is the solution to $2x + 9 = 5x - 6$?
A) 3
B) 5
C) 9
D) 15

MATH3-M1-002
Prompt:
A taxi ride costs $4.50 plus $2.25 for each mile traveled. What is the total cost of a taxi ride that is 6 miles long?
A) $13.50
B) $16.00
C) $18.00
D) $22.50

MATH3-M1-003
Prompt:
The numbers of pages read by 6 students in one day are 18, 22, 18, 25, 31, and 18. What is the mode of this data set?
A) 18
B) 22
C) 25
D) 31

MATH3-M1-004
Prompt:
The equation $y = -4x + 12$ represents a line in the xy-plane. What is the y-coordinate of the point where the line crosses the y-axis?
A) −4
B) 0
C) 4
D) 12

MATH3-M1-005
Prompt:
A ribbon is 7 yards long. If 1 yard equals 3 feet, how many feet long is the ribbon?
A) 10
B) 14
C) 21
D) 28

MATH3-M1-006
Prompt:
What is the solution (x, y) to the system of equations?
$x = y + 4$
$2x + y = 19$
A) (5, 9)

B) (7, 3)
C) (8, 4)
D) (9, 5)

MATH3-M1-007
Prompt:
Which inequality is equivalent to $6 - 2x > 14$?
A) $x < -4$
B) $x > -4$
C) $x < 4$
D) $x > 4$

MATH3-M1-008
Prompt:
For a positive integer n, which expression is equivalent to $(n^3)^2$?
A) n^5
B) n^6
C) n^9
D) $2n^3$

MATH3-M1-009
Prompt:
Which expression is equivalent to $6m + 9$?
A) $3(2m + 3)$
B) $3(2m + 9)$
C) $6(m + 9)$
D) $9(m + 6)$

MATH3-M1-010
Prompt:
A spinner is divided into 10 equal sections numbered 1 through 10. If the spinner is spun once, what is the probability that it lands on a number greater than 7?
A) 1/10
B) 3/10
C) 7/10
D) 3/7

MATH3-M1-011
Prompt:
A parallelogram has a base of 11 centimeters and a height of 5 centimeters. What is the area, in square centimeters, of the parallelogram?
A) 16
B) 27.5
C) 55
D) 110

MATH3-M1-012
Prompt:
In the xy-plane, points A and B have coordinates $(-3, 2)$ and $(4, 2)$, respectively. What is the distance between A and B?
A) 1

B) 5
C) 7
D) 9

MATH3-M1-013
Prompt:
What is the value of $2^2 + 3^2$?
A) 10
B) 12
C) 13
D) 25

MATH3-M1-014
Prompt:
If $q(x) = x^2 + 2x$, what is $q(5)$?
A) 25
B) 30
C) 35
D) 45

MATH3-M1-015
Prompt:
A map uses a scale of 1 inch to represent 6 miles. On the map, the distance between two towns is 4.5 inches. What is the actual distance, in miles, between the two towns?
A) 10.5
B) 24
C) 27
D) 30.5

MATH3-M1-016
Prompt:
The table shows the number of cups of water remaining in a pitcher after different numbers of cups are poured out.

Cups poured out: 0, 1, 2, 3
Cups remaining: 12, 10, 8, 6

If the pattern continues, how many cups of water remain after 5 cups are poured out?
A) 0
B) 2
C) 4
D) 5

MATH3-M1-017
Prompt:
A rectangular garden has an area of 96 square feet. The width of the garden is 8 feet. What is the length, in feet, of the garden?
A) 10
B) 12
C) 16
D) 24

MATH3-M1-018
Prompt:
Which value is equal to $\sqrt{81} - \sqrt[3]{27}$?
A) 3
B) 6
C) 9
D) 12

MATH3-M1-019
Prompt:
A straight angle is divided into two angles. One angle measures 37°. What is the measure, in degrees, of the other angle?
A) 53
B) 90
C) 143
D) 153

MATH3-M1-020
Prompt:
A line in the xy-plane is represented by $2x + y = 9$. Which ordered pair lies on this line?
A) (1, 5)
B) (2, 4)
C) (3, 3)
D) (4, 2)

MATH3-M1-021
Prompt:
A rectangular box has a length of 10 inches, a width of 3 inches, and a volume of 180 cubic inches. What is the height, in inches, of the box?
A) 3
B) 6
C) 9
D) 12

MATH3-M1-022
Prompt:
A group of 40 students was asked whether they prefer drama, art, or music. The results are shown.

Preference: Drama, Art, Music
Number of students: 9, 14, 17

What percent of the students prefer art?
A) 14%
B) 28%
C) 35%
D) 40%

PART B — MODULE 2 QUESTIONS

MATH3-M2-001
Prompt:

What is the solution to $(3x + 8)/2 = x + 11$?
A) 10
B) 12
C) 14
D) 18

MATH3-M2-002
Prompt:
A line passes through the points $(-1, 4)$ and $(5, 16)$. Which equation represents the line?
A) $y = 2x + 6$
B) $y = 2x - 6$
C) $y = 3x + 7$
D) $y = 4x + 8$

MATH3-M2-003
Prompt:
If $(x - 4)^2 = 49$, which of the following is a possible value of x?
A) -11
B) -3
C) 4
D) 11

MATH3-M2-004
Prompt:
Which expression is equivalent to $2x^2 + 7x - 15$?
A) $(2x - 3)(x + 5)$
B) $(2x + 3)(x - 5)$
C) $(x - 3)(2x + 5)$
D) $(x + 3)(2x - 5)$

MATH3-M2-005
Prompt:
The function r is defined by $r(x) = 5x^2 - 20x + 18$. Which expression shows r(x) in a form that displays the vertex of its graph?
A) $5(x - 2)^2 - 2$
B) $5(x + 2)^2 - 2$
C) $5(x - 2)^2 + 18$
D) $5(x + 2)^2 + 18$

MATH3-M2-006
Prompt:
A store buys a lamp for $36 and sells it for 30% more than the price the store paid. What is the selling price of the lamp?
A) $40.80
B) $46.80
C) $48.00
D) $66.00

MATH3-M2-007
Prompt:
The table shows the cost, in dollars, to rent a kayak for different numbers of hours.

Hours: 1, 2, 4, 6
Cost: 18, 26, 42, 58

The cost is a linear function of the number of hours. What is the cost, in dollars, to rent the kayak for 9 hours?
A) 72
B) 78
C) 82
D) 90

MATH3-M2-008
Prompt:
A triangle has side lengths 9, 12, and 15. What is the area, in square units, of the triangle?
A) 36
B) 54
C) 67.5
D) 108

MATH3-M2-009
Prompt:
In a right triangle, $\cos \theta = 7/25$. If the hypotenuse has length 50, what is the length of the side adjacent to angle θ?
A) 7
B) 14
C) 25
D) 48

MATH3-M2-010
Prompt:
The equation $3y - 6x = 15$ represents a line. What is the x-coordinate of the x-intercept of the line?
A) $-5/2$
B) 0
C) 5/2
D) 5

MATH3-M2-011
Prompt:
The function f is defined by $f(x) = 2x - 7$. If $f(a) = 3a + 5$, what is the value of a?
A) -12
B) -8
C) 8
D) 12

MATH3-M2-012
Prompt:
Which expression is equivalent to $(12x^3y^2)/(4xy)$, where $x \neq 0$ and $y \neq 0$?
A) $3x^2y$
B) $3x^4y^3$
C) $8x^2y$
D) $8x^3y^2$

MATH3-M2-013
Prompt:
A circle has center $(-1, 4)$. The point $(2, 8)$ lies on the circle. What is the radius of the circle?
A) 3
B) 4
C) 5
D) 7

MATH3-M2-014
Prompt:
For what value of t does the system of equations have no solution?
$y = -2x + 7$
$4x + 2y = t$
A) 0
B) 7
C) 10
D) 14

MATH3-M2-015
Prompt:
The table shows values of the function f.

x: 1, 2, 3, 4
f(x): 5, 9, 15, 23

Which equation defines f?
A) $f(x) = x^2 + 4$
B) $f(x) = x^2 + x + 3$
C) $f(x) = 2x^2 + 3$
D) $f(x) = 3x + 2$

MATH3-M2-016
Prompt:
What is the solution to $4/(x - 1) = 1/3$?
A) 5
B) 7
C) 12
D) 13

MATH3-M2-017
Prompt:
A sample contains 240 grams of a solution that is 15% salt by mass. How many grams of salt are in the solution?
A) 24
B) 30
C) 36
D) 40

MATH3-M2-018
Prompt:
A cone has a radius of 3 centimeters and a height of 8 centimeters. The volume of a cone is V =

$(1/3)\pi r^2h$. What is the volume, in cubic centimeters, of the cone?
A) 24π
B) 36π
C) 72π
D) 216π

MATH3-M2-019
Prompt:
For constants a and b, the expression $ax + b$ is equivalent to $5(2x − 3) − 4x$. What is the value of $a + b$?
A) −9
B) −6
C) 6
D) 21

MATH3-M2-020
Prompt:
A data set has 9 values. The mean of the data set is 18. If one value, 30, is removed from the data set, what is the mean of the remaining 8 values?
A) 15.5
B) 16.5
C) 17
D) 18

MATH3-M2-021
Prompt:
The expression $(x + 2)^2 − 16$ is equivalent to $x(x + k)$ for all values of x. What is the value of k?
A) −12
B) −4
C) 4
D) 8

MATH3-M2-022
Prompt:
A company's total cost C, in dollars, to make n items is modeled by $C = 250 + 6n$. The company sells each item for $11. For what value of n will the total revenue equal the total cost?
A) 25
B) 40
C) 50
D) 60

PART C — MODULE 1 ANSWER KEY + EXPLANATIONS

MATH3-M1-001 — Correct: 5 — Explanation: Subtract 2x from both sides to get $9 = 3x − 6$. Add 6 to get $15 = 3x$, so $x = 5$.
MATH3-M1-002 — Correct: $18.00 — Explanation: The total cost is $4.50 + 2.25(6) = 4.50 + 13.50 = 18.00$.
MATH3-M1-003 — Correct: 18 — Explanation: The mode is the value that appears most often. The number 18 appears three times.
MATH3-M1-004 — Correct: 12 — Explanation: In $y = −4x + 12$, the line crosses the y-axis when $x = 0$, giving $y = 12$.

MATH3-M1-005 — Correct: 21 — Explanation: Since 1 yard equals 3 feet, 7 yards equals $7 \times 3 = 21$ feet.
MATH3-M1-006 — Correct: (9, 5) — Explanation: Substitute $x = y + 4$ into $2x + y = 19$ to get $2(y + 4) + y = 19$. Then $3y + 8 = 19$, so $y = 5$ and $x = 9$.
MATH3-M1-007 — Correct: $x < -4$ — Explanation: Subtract 6 to get $-2x > 8$. Dividing by -2 reverses the inequality, so $x < -4$.
MATH3-M1-008 — Correct: n^6 — Explanation: Raising a power to a power means multiplying exponents: $(n^3)^2 = n^6$.
MATH3-M1-009 — Correct: $3(2m + 3)$ — Explanation: The greatest common factor of 6m and 9 is 3, so $6m + 9 = 3(2m + 3)$.
MATH3-M1-010 — Correct: 3/10 — Explanation: The numbers greater than 7 are 8, 9, and 10, so 3 of the 10 equal sections are favorable.
MATH3-M1-011 — Correct: 55 — Explanation: The area of a parallelogram is base × height, so the area is $11 \times 5 = 55$.
MATH3-M1-012 — Correct: 7 — Explanation: The points have the same y-coordinate, so the distance is the difference in x-coordinates: $4 - (-3) = 7$.
MATH3-M1-013 — Correct: 13 — Explanation: $2^2 + 3^2 = 4 + 9 = 13$.
MATH3-M1-014 — Correct: 35 — Explanation: Substitute 5 for x: $q(5) = 5^2 + 2(5) = 25 + 10 = 35$.
MATH3-M1-015 — Correct: 27 — Explanation: Each inch represents 6 miles, so 4.5 inches represents $4.5 \times 6 = 27$ miles.
MATH3-M1-016 — Correct: 2 — Explanation: The number of cups remaining decreases by 2 for each cup poured out. After 5 cups are poured out, $12 - 2(5) = 2$ cups remain.
MATH3-M1-017 — Correct: 12 — Explanation: Area equals length × width, so 96 = length × 8. The length is $96 \div 8 = 12$.
MATH3-M1-018 — Correct: 6 — Explanation: $\sqrt{81} = 9$ and $\sqrt[3]{27} = 3$, so $\sqrt{81} - \sqrt[3]{27} = 6$.
MATH3-M1-019 — Correct: 143 — Explanation: A straight angle measures 180°, so the other angle is $180 - 37 = 143°$.
MATH3-M1-020 — Correct: (3, 3) — Explanation: Substituting (3, 3) gives $2(3) + 3 = 9$, which is true.
MATH3-M1-021 — Correct: 6 — Explanation: The volume is length × width × height, so $180 = 10 \times 3 \times h$. Thus $h = 6$.
MATH3-M1-022 — Correct: 35% — Explanation: The fraction that prefer art is $14/40 = 0.35$, which is 35%.

PART D — MODULE 2 ANSWER KEY + EXPLANATIONS

MATH3-M2-001 — Correct: 14 — Explanation: Multiply both sides by 2 to get $3x + 8 = 2x + 22$. Subtract 2x and 8 to get $x = 14$.
MATH3-M2-002 — Correct: $y = 2x + 6$ — Explanation: The slope is $(16 - 4)/(5 - (-1)) = 12/6 = 2$. Using point $(-1, 4)$, $4 = 2(-1) + b$, so $b = 6$.
MATH3-M2-003 — Correct: 11 — Explanation: From $(x - 4)^2 = 49$, $x - 4 = 7$ or $x - 4 = -7$. Thus $x = 11$ or $x = -3$.
MATH3-M2-004 — Correct: $(2x - 3)(x + 5)$ — Explanation: Expanding $(2x - 3)(x + 5)$ gives $2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$.
MATH3-M2-005 — Correct: $5(x - 2)^2 - 2$ — Explanation: Factor 5 from the quadratic and linear terms: $5(x^2 - 4x) + 18 = 5((x - 2)^2 - 4) + 18 = 5(x - 2)^2 - 2$.
MATH3-M2-006 — Correct: \$46.80 — Explanation: A 30% increase on \$36 is $0.30 \times 36 = \$10.80$, so the selling price is $36 + 10.80 = \$46.80$.
MATH3-M2-007 — Correct: 82 — Explanation: The cost increases by \$8 for each additional hour. The cost is $10 + 8h$, so for 9 hours the cost is $10 + 8(9) = 82$.
MATH3-M2-008 — Correct: 54 — Explanation: Since $9^2 + 12^2 = 15^2$, the triangle is right. Its area

is $(1/2)(9)(12) = 54$.
MATH3-M2-009 — Correct: 14 — Explanation: Cosine is adjacent/hypotenuse. Since $7/25 = \text{adjacent}/50$, the adjacent side length is $50 \times 7/25 = 14$.
MATH3-M2-010 — Correct: $-5/2$ — Explanation: The x-intercept occurs when $y = 0$. Substituting gives $-6x = 15$, so $x = -15/6 = -5/2$.
MATH3-M2-011 — Correct: -12 — Explanation: Since $f(a) = 2a - 7$, set $2a - 7 = 3a + 5$. Solving gives $-12 = a$.
MATH3-M2-012 — Correct: $3x^2y$ — Explanation: Divide coefficients and subtract exponents: $(12x^3y^2)/(4xy) = 3x^2y$.
MATH3-M2-013 — Correct: 5 — Explanation: The radius is the distance from $(-1, 4)$ to $(2, 8)$: $\sqrt{(2 - (-1))^2 + (8 - 4)^2} = \sqrt{3^2 + 4^2} = 5$.
MATH3-M2-014 — Correct: 10 — Explanation: From $y = -2x + 7$, multiplying by 2 gives $2y = -4x + 14$, so $4x + 2y = 14$. The system has no solution when t is any value other than 14; among the choices, 10 works.
MATH3-M2-015 — Correct: $f(x) = x^2 + x + 3$ — Explanation: Substituting $x = 1, 2, 3$, and 4 into $x^2 + x + 3$ gives 5, 9, 15, and 23, matching the table.
MATH3-M2-016 — Correct: 13 — Explanation: Cross-multiply to get $12 = x - 1$, so $x = 13$.
MATH3-M2-017 — Correct: 36 — Explanation: 15% of 240 is $0.15 \times 240 = 36$ grams.
MATH3-M2-018 — Correct: 24π — Explanation: Using $V = (1/3)\pi r^2h$ with $r = 3$ and $h = 8$ gives $V = (1/3)\pi(9)(8) = 24\pi$.
MATH3-M2-019 — Correct: -9 — Explanation: Simplify $5(2x - 3) - 4x$ to get $10x - 15 - 4x = 6x - 15$. Thus $a = 6$ and $b = -15$, so $a + b = -9$.
MATH3-M2-020 — Correct: 16.5 — Explanation: The original total is $9 \times 18 = 162$. After removing 30, the remaining total is 132. The new mean is $132 \div 8 = 16.5$.
MATH3-M2-021 — Correct: 4 — Explanation: $(x + 2)^2 - 16 = x^2 + 4x + 4 - 16 = x^2 + 4x - 12$. Since this equals $x(x + k) = x^2 + kx$, $k = 4$.
MATH3-M2-022 — Correct: 50 — Explanation: Revenue from selling n items is $11n$. Set $11n = 250 + 6n$, giving $5n = 250$, so $n = 50$.